IntelliBed: A Load Cell-Based Sleep Monitoring System

Vikram Singh
University of Massachusetts Amherst, USA
vdsingh@umass.edu

Ryan Baker
University of Massachusetts Amherst, USA
rwbaker@umass.edu

ABSTRACT
Various studies show that sleep is essential for a variety of bodily functions. These include brain function (memory, learning, creativity, problem-solving skills, and focus) and body function (including muscle repair, tissue growth, restoring energy, immunity, blood pressure, and weight gain). According to the CDC, more than 70 million Americans suffer from Chronic sleep problems, many of whom avoid facing these issues because they aren't aware of them or aren't sure where to start. We have developed a system called IntelliBed that uses Load Cell Sensors to unobtrusively collect weight distribution data throughout the night. It then uses Machine Learning to classify sleep events based on this data in order to measure restlessness. This system can be used to support research studies by providing a new type of data to analyze when looking into causes and solutions of sleeping disorders. It can also be used to measure things like the impact of different time periods or environmental events on sleep. For instance, students may have lower quality sleep, on average, during final exam periods. The applicability of load cell sensors (and weight distribution data in general) for sleep monitoring is underexplored, with many current systems using obtrusive wearable sensors (like smart watches) to accomplish this task. While these devices have their advantages (accuracy and simplicity to name a few), it is possible that such wearable devices could directly impact sleep quality (and thus data) by causing discomfort during sleep.

Author Keywords
Sleep monitoring; load cells; sensors; machine learning; KNN;

1) INTRODUCTION

1.1) The Problem and our Contribution

Sleep disorders like insomnia and sleep apnea affect over 70 million Americans every night, leading to a variety of health problems including everything from weight gain to heart disease. There has been a large amount of research and development in (a) the interest of tracking/monitoring sleep (b) the interest of finding solutions to these issues, and (c) explaining why they occur to begin with. Many of the greatest challenges and much of the importance of general research in the field stem from (a) since the issues that are explored within (b) and (c) involve analysis questions which first require exploring (a) in order to acquire data. IntelliBed is intended to be a system that contributes to (a) for the sake of (b) and (c). The system may also be applicable to users who are interested in tracking their sleep for any reason they see fit, but it is important to note that the system and its mobile application interface are not intended to directly introduce any explanation or logical reasoning behind how the sensor data maps to sleep quality/disorders. In other words, we don't offer a comprehensive diagnosis/analysis of the data from a psychological perspective.

1.2) Existing Solutions and Pros/Cons of IntelliBed

Many researchers interested in this issue have used advanced methods like a ballistocardiogram, polysomnography, or photoplethysmogram. Many of these methods, however, involve the use of physiological sensors to collect data, which can be obtrusive, affecting the data by impacting sleep. Many researchers have also used off-the-shelf actigraphy based systems like Fitbit, in order to collect accelerometer data that provides information about sleep movement. These can also be intrusive, causing the same problem as previously mentioned. These systems do certainly have their advantages, however, mostly in the realm of simplicity and accuracy. Polysomnographies, for instance, provide fine grained data, yielding very accurate results, and are the leading method in clinical use (often used for diagnosing insomnia). The space of using load cell sensors/weight distribution data for tasks such as classifying sleep events is underexplored. These kinds of systems are much more cost effective and efficient than aforementioned systems, and we intend for IntelliBed to be used in studies where technology for advanced methods is not available or where researchers are interested in exploring sleep monitoring with non-wearable sensors.

1.3) A Brief Description of of IntelliBed

IntelliBed is a Raspberry Pi 2 system that uses a total of eight 50kg load cell sensors (with two under each leg of the bed). It stores time synchronized weight data for each leg of the bed as a time series. We take the raw weight data for

each leg, and extract the following features: . We then detect general movement and events like rollovers from a trained KNN classifier on these features. From here, we export windowed feature data to Google Cloud's Realtime Database, which is interfaced by an iOS application. The iOS application mainly uses Swift 5, UIKit, and Swift-LineChart to visualize and present the data so that it is available for analysis.

2) RELATED WORK / LITERATURE REVIEW

2.1) *You Are How You Sleep: Personalized Sleep Monitoring Based on Wrist Temperature and Accelerometer Data* by Jing Wei and Jennifer Boger

Researchers who wrote this journal explore the usage of wrist temperature and actigraphy during sleep to "investigate the integration of circadian rhythms into sleep evaluation". Their system involves a wearable device with a custom-built temperature and accelerometry wristband. They measure the wrist temperature for early and late sleepers and compare the first wrist temperature onset times. With the help of 10 participants, researchers were able to categorize wrist temperatures to be day time or night time temperatures by looking at the timestamps for sleep onset and offset. Researchers also explore an issue involving external factors and explain how their algorithm accounts for it. Factors such as ambient temperature can cause unavoidable changes in wrist temperature. In order to account for this, their algorithm looks at the onset and offset time of past data, mitigating changes from outside factors.

The system explained in this paper is very similar to various existing systems, in that it is wearable and uses accelerometer data to monitor sleep. Their use of wrist temperature monitoring, however, is a relatively new and innovative idea. As previously mentioned, they ran into the issue of dealing with extraneous factors such as ambient temperature. This is not a problem for IntelliBed because the main condition that we must guarantee is that all sensors are in a similar condition to one another. It may be true that things like the temperature can slightly affect the load cell sensors under the leg extenders, but this is not an issue because if all of the sensors are affected (and they will be), the system will still be calibrated correctly.

2.2) *IoT-Based Wireless Polysomnography Intelligent System for Sleep Monitoring* by Chin-Teng Lin, Mukesh Prasad, Chia-Hsin Chung, Deepak Puthal, Hesham El-Sayed, Sharmi Sankar, Yu-Kai Wang, Jagendra Singh, Arun Kumar Sangaiah

This journal explores the use of polysomnography - the gold standard for diagnosing obstructive sleep apnea. The use of this system generally requires participants to sleep overnight in a laboratory, as researchers mention, this can disturb sleep and result in an incomplete or inaccurate test. These researchers explain the creation of a wireless polysomnography system that is usable in one's own home. Their system is battery-powered, miniature, and portable, making it easy to set up. This system also has a lower cost, but has a higher proportion of inadequate examinations. This system looks at specific stages of sleep and observes how long these last each night. These stages include: awakening, REM, N1, N2, and N3. Their system had a low relative error for awakening and REM (3%), a slightly higher relative error for N1 and N2 (7% and 9%), and a high relative error for N3 (51%). The reason for the high error for N3 was that "Stage N2 may be interpreted as Stage N3 if some segments of the physiology signals were extremely similar".

While the system explored in this paper has a lower cost than traditional polysomnography systems, it is still quite expensive, and unobtainable for many people. It is also still obtrusive, as it is still polysomnography and requires electrodes to be attached to the participant who is being monitored. That said, this system is difficult to compare with IntelliBed. This system monitors and identifies specific stages of sleep, as noted above. As a result, different challenges arise and different results are drawn. IntelliBed, on the other hand, accomplishes the task of identifying sleep events. IntelliBed could be used in conjunction with a polysomnography to observe active behaviors during different stages of sleep. For instance, a person may have more sleep events during REM sleep, which may say something about their condition.

3) USER STUDY DESIGN AND DATA COLLECTION/PROCESSING

Our dataset was collected at the University of Massachusetts Amherst in a standard dorm by using a standard UMass twin XL dorm bed. We had a total of two participants perform the four total actions associated with our model. With each action to be measured, the system was first calibrated using two 30 pound dumbbells placed in the center of the bed. Then, the user climbed onto the bed, the other started the script to collect data, and the user present on the bed repeated the specified action around 30-40 times. Once finished, the data collection script was terminated, and the user climbed off of the bed.

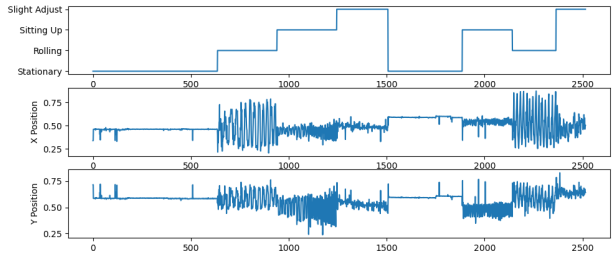


Figure 1: This is a graph of the raw x and y positions along with the ground truth class associated with it. It can be seen that when stationary, the x and y positions remain constant. When rolling, the x positions decrease or increase depending on the direction rolled. When sitting up, the y values would decrease, and the x values did not significantly change. When slightly adjusting, both x and y values appear similar to stationary, but with more variance.

Processing the data / Feature Extraction

We windowed the raw data with a window size of 32 and window step of 4. We extracted the following features for each set of X and Y: Mean, Median, Standard Deviation, Variance, Slope, and Root Mean Squared. We did not use any of the raw sensor values because with the current system, it cannot be guaranteed that the values will have the same scalars for every session. Furthermore, it should not need to take into account the person's weight, we are mainly looking at the relative position. It is also very redundant, since we are not interested in weight

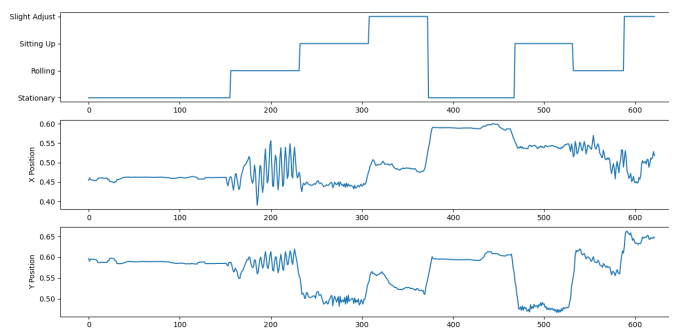


Figure 3: This is a graph of the mean of x and y positions. This appears similar to the raw data, but with less variance.

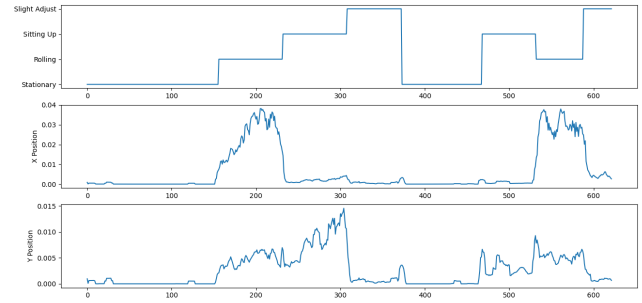


Figure 2: This is a graph of the variance of x and y positions. When stationary, both variances are very low. When rolling, the x variance is significantly higher than the y variance. When sitting up, the y variance is significantly higher than the x variance. Variance doesn't provide any way to differentiate between stationary and slight adjustments however

We did not use peaks, or any frequency domain features, as we felt that those are irrelevant for this application. For instance, since one roll would take up the majority of a window, there is no frequency, it would look linear or parabolic. The same applies to all other actions as well

We then did k-Fold Cross-Validation using a KNN model, in which we created 5 equal sized groups of windows. For each window we trained a KNN (number of neighbors = 10) on the other 4 groups and tested it on that window that had not been trained with. We found that the average F1 score on the test data was around 0.98-0.99. This seems promising, but since our sample size is not very large, undependable. Adding more data from different participants would likely lower our accuracy due to variety. It would also reduce overfitting, and allow the model to better predict on new people.

	Stationary	Roll	Situp	Adjust
Stationary	46	0	0	0
Roll	0	30	0	0
Sit Up	0	0	28	0
Adjust	0	1	0	19

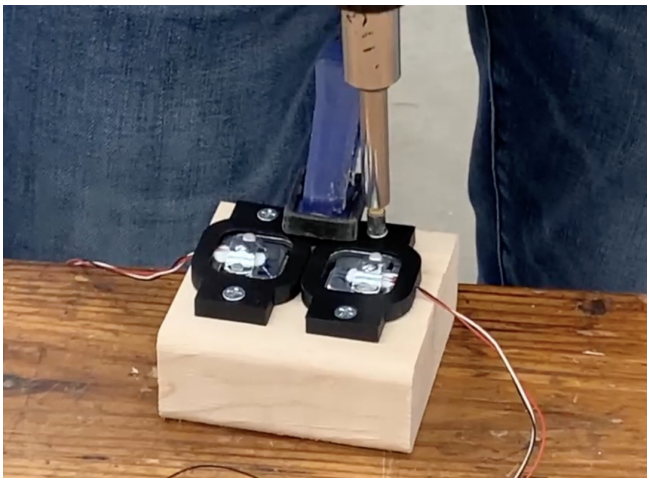
Figure 4: A confusion matrix showing how well our classifier predicts sleep events.

4) SYSTEM DESCRIPTION

4.1) Hardware/Other Resources

Our system uses the following hardware/resources:

- 4 HX711s: these serve as amplifiers to the load cell
- 1 Raspberry Pi 2
- 8 50kg load cell sensors
- Other electrical components (breadboards, wires, etc.)
- 4 wooden blocks



This is of the leg extenders that goes under the leg of a bed frame.

4.2) Software/Services

Our system uses the following software/services:

- Python 3 (Programming Language): to write the code for the Raspberry Pi 2, which collects and analyzes the data
  - Using a developed HX711 library which can be found here: https://github.com/tatobari/hx711py
- Google Cloud Firebase Realtime Database (Service): write and store the processed data from the raspberry pi, read the processed data from the iOS application interface.
- Swift 5 (Programming Language)/Xcode: to build the application that interfaces the realtime database and visualize the processed information
  - UIKit + Swift-LineChart: libraries to work with the user interface and graphs.
  - Swift-LineChart: https://github.com/zemirco/swift-linechart
- The code for the system can be found here: https://github.com/rwbakerUMASS/IntelliBed-System
- The code for the mobile application can be found here: https://github.com/vdsingh/IntelliBed-iOS

4.3) System Architecture

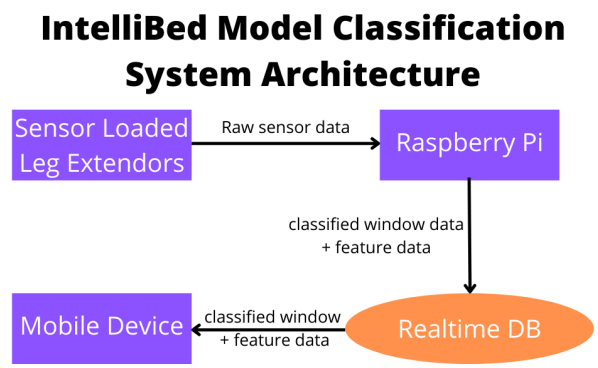


Figure 5: IntelliBed Model Classification System Architecture. This diagram shows a basic overview of the components of IntelliBed and how they work together to classify data.

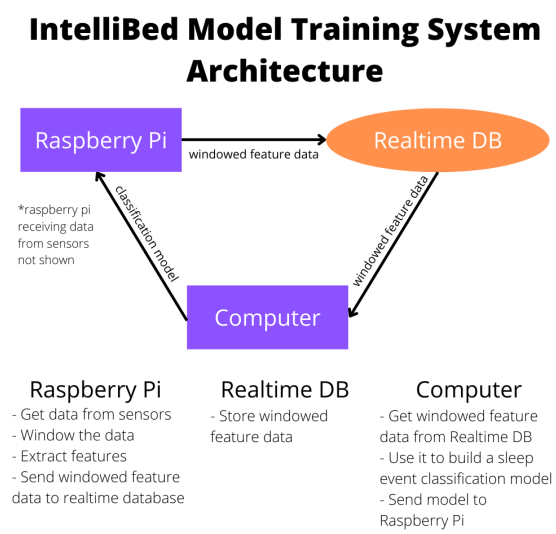


Figure 6: IntelliBed Model Training System Architecture. This diagram shows a basic overview of the components of IntelliBed that are required to train our system.

By using these components in conjunction, we have built an efficient and easy-to-use system for collecting, storing, and processing data, along with building, delivering, and using a classification model. The process can be broken down into two steps: training and classifying.

Training: As can be seen by the system architecture map above (figure 2), our model will exist on the raspberry pi (which is the brain of the main IntelliBed system). Producing this model starts with data collection. Raw weight data from each leg is collected from the sensor loaded leg extenders and transmitted to the Raspberry Pi. With Python, the Raspberry Pi windows the data (windows of 32 data points), allowing us to work with time series data, and extracts the following features from x and y value data: mean, median, standard deviation, variance, root mean squared, and slope. It then sends this windowed feature data to a realtime database hosted by Google Cloud.

5) INTERFACING THE DATA WITH A MOBILE (IOS) APPLICATION

Having data for sleep events and sleep scores is pointless unless users and participants have access to an interface that can effectively and efficiently convey the data. The IntelliBed System stores the data in Google Cloud's Firebase Realtime Database with a Python script. The same database is then interfaced by an iOS application (programmed with Swift 5) that uses the Swfit-LineChart by Mirco Zeiss (which can be found here: https://github.com/zemirco/swift-linechart) library to visualize the data. This gives users the ability to see the data behind sleep, revealing what leads to the classification that a window is given. The application first gets all of the data from the database and windows it (32 data points per window). Users are first met with a table view screen that shows all of the windows available. They can then tap on a cell in the table view to see more about that window. The information available pertains to the basic features of our classification model: mean, median, variance, standard deviation. We have plans to implement root mean square and slope in the future, as those implementations have proven to be difficult with swift. Along with the feature information, users are also presented with two chart visualizations, which show time series data for both X and Y position in that window. From this, users get a visual representation of the data, allowing them to further understand the classification of that data. Finally, the user can see the actual classification so that they may understand the correspondence between the data and the class.

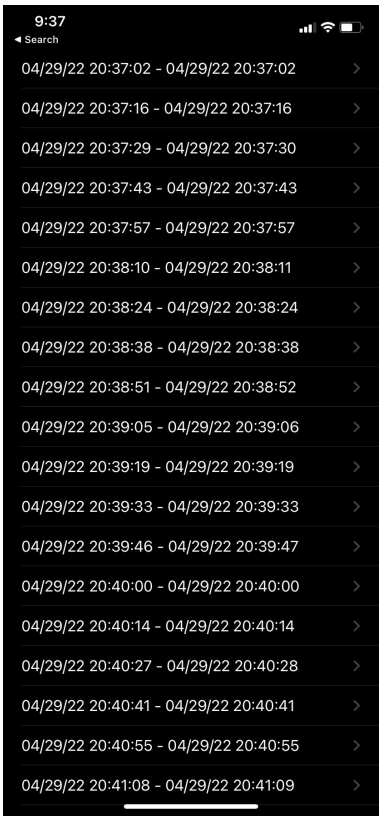


Figure 7: The window selection. This screen offers a table view of all data windows available on the realtime database. Each cell in the table view is labeled according to the timestamps of the first and last data points in the window. By clicking on a cell, the user will be taken to the window information view page as denoted by Figure 8.

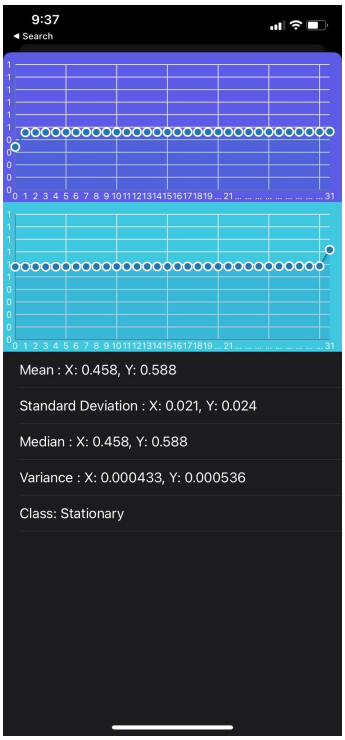


Figure 8: The window information view page. This screen shows a visualization of the change in x (top chart) and y values (bottom chart) of the data points included in a given window. In the table view underneath, the basic features calculated with this window are shown. These include mean, standard deviation, median, and variance.

7) PROBLEMS WE FACED

7.1) Capacity Limits

Since we are building a system for sleep monitoring that does not involve a wearable sensor, we had to give some thought to how much weight we need to handle. We wanted to account for individuals who are 200 pounds maximum. We also had to account for the weight of the wooden/steel bed frame, the mattress, and whatever else might be on the bed. Thus, we wanted to account for about 400 pounds total, to be on the safe side. Each load cell has a capacity limit of 50kg, so if we had used 4 load cells, we would have a limit of 200kg or 440 pounds. While this is greater than the 400 pounds that we planned to account for, it should be noted that a person is generally near the top of their bed when sleeping or doing most other activities, thus if we only count 2 load cells worth of capacity, we can sustain 100kg, or about 220 pounds. This is likely not enough for a person who is 200 pounds, after we factor in the weight of everything else. To solve this problem, we bought a total of 8 load cells, and under each leg, we used pairs of 2 in order to distribute the load.

7.2) Drifting Data

While building the IntelliBed system, one common issue we faced was drifting data. When we connected our load cell sensors to the raspberry pi and observed the values from the sensors, the values would steadily increase even when no force was being applied. We had a couple of guesses as to why this occurred, including: an unsteady power supply, faultiness of the HX711s (hardware issues), electromagnetic interference, or a faulty breadboard. By spending some time diagnosing the issue, we found that moving the wire connected to the load cell sensors caused the values to change dramatically. After working to solve this issue for a couple days, we realized that there was an issue with the breadboard that we were using. We had previously used wires that were thicker, which changed the breadboard. When we began to use thinner wires, the breadboard was slightly less responsive to them. We fixed this problem by simply using a new breadboard.

8) ADVANTAGES OF OUR SYSTEM

8.1) Comfortability and Affordability

One of the main disadvantages to actigraphy systems is that they involve a wearable sensor (usually in the form of a smart watch such as a fitbit). There are several issues with these that IntelliBed avoids. Firstly, these wearable sensors can be quite uncomfortable during sleep, potentially causing wrist pain. As a result, these can affect sleep quality, which affects our data, leading to potentially inaccurate results. We want to know if people are restless in their sleep because of a sleeping disorder, not due to an uncomfortable sensor.

Secondly, typical systems for monitoring sleep such as ballistocardiography, polysomnography, or photoplethysmogram are very expensive, and unobtainable for many people. Many of these systems also require participants to sleep overnight in a laboratory (also potentially causing sleep discomfort). Even actigraphy devices can be quite expensive, with many devices costing well over $100. IntelliBed was built with a budget of $56. This includes $30 for the Raspberry Pi 2 Model B, $16 for 8 load cell sensors, and $10 for miscellaneous electronics such as breadboards and wires. Thus, this system is very attainable for researchers or other consumers who may be interested in its uses, and can be used in a person's own home.

8.2) Data Perspective

As previously mentioned, the use of weight distribution data to classify sleep events is a topic that is widely

unexplored. Using this kind of technology to identify sleep events offers data from an entirely new perspective, allowing it to be used to increase support and credibility of already existing studies. For instance, if a person is being tested for sleep apnea or insomnia, using this system can help researchers understand what is occurring from a completely different point of view: in terms of their distinct actions during sleep.

9) DRAWBACKS OF OUR SYSTEM

9.1) Bed Size and Frame

The data that our model is trained and tested on was collected by using a standard UMass twin XL dorm bed and frame. Our model may not be as accurate if a different sized bed is used, and may be rendered useless if a bed frame is not present. Our system acts on the assumption that the entirety of the weight present that is being measured is distributed across the leg extenders. Without a bed frame (with legs), there will be weight that is not accounted for (in the event that the mattress is touching the ground), which may affect the model. While our model is not directly founded on the dimensions of the bed, a larger bed gives participants more room to move and for sleep events. As a result, the sleep events may occur differently both in terms of the data and in terms of the movement itself. For instance, rolling over on a twin size bed may be different from rolling over on a king size because some adjustment may be necessary on the twin to avoid falling off.

9.2) Number of Users

Our system is only useful for monitoring users who sleep alone - it is not built for more than one person on the bed. Since we calculate the x and y position of the participant by triangulation based on leg extender weight measurements, multiple people present on the bed will cause the x and y values from the triangulation to be inaccurate due to the leg extender weight measurements being inaccurate (with respect to a singular person). In this regard, actigraphy based systems will be more applicable and useful, since they can monitor the sleep of a single person regardless of the number of people present.

10) CONCLUSION

In this paper, we have demonstrated how weight distribution data during a person's sleep can be used to predict and classify sleep events. The classification of such events can be used by researchers as supporting evidence of information related to the causes and effects of sleep-related medical conditions. While it is true that state of the art systems like ballistocardiography and polysomnography may be more accurate and precise, these systems are often expensive, obtrusive, and unobtainable. We have shown that a cheaper and more obtainable alternative is possible and that it offers a similar perspective on sleep data to actigraphy, but without being wearable. IntelliBed shows considerable promise for monitoring and potentially developing holistic models of the complex relationships between sleep quality and sleep events.

To summarize, the various components of IntelliBed are: the leg extenders and hardware, which are necessary to collect data, the raspberry pi and computer, which are necessary to retrieve the data and train/invoke the KNN model, the realtime database, which stores the data on the cloud, and the iOS interface, which is used to visualize the data for each window.

REFERENCES

1. Wei, Jing, and Jennifer Boger. "You Are How You Sleep: Personalized Sleep Monitoring Based on Wrist Temperature and Accelerometer Data." *Proceedings of the 13th EAI International Conference on Pervasive Computing Technologies for Healthcare - Demos and Posters*, 2019, https://doi.org/10.4108/eai.20-5-2019.2282879.

2. C. -T. Lin *et al*., "IoT-Based Wireless Polysomnography Intelligent System for Sleep Monitoring," in *IEEE Access*, vol. 6, pp. 405-414, 2018, doi: 10.1109/ACCESS.2017.2765702.